Modeling Gas Exchange in the Lungs

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Background

- Gas exchange facilitates many vital bodily functions (metabolism)
- More than 34 million Americans suffer from chronic lung diseases[1]
- Understanding of gas exchange can inform decisions in clinical setting

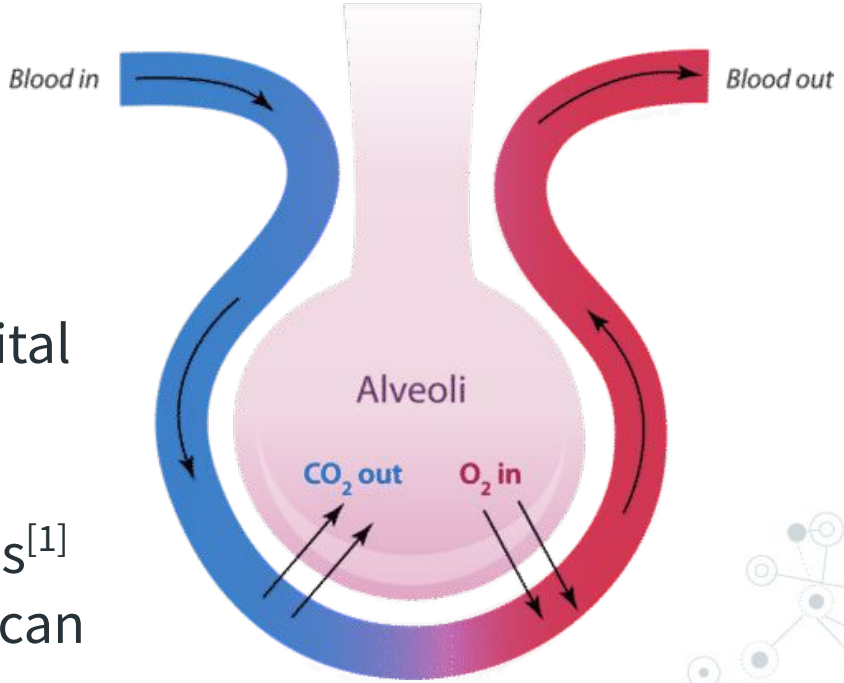
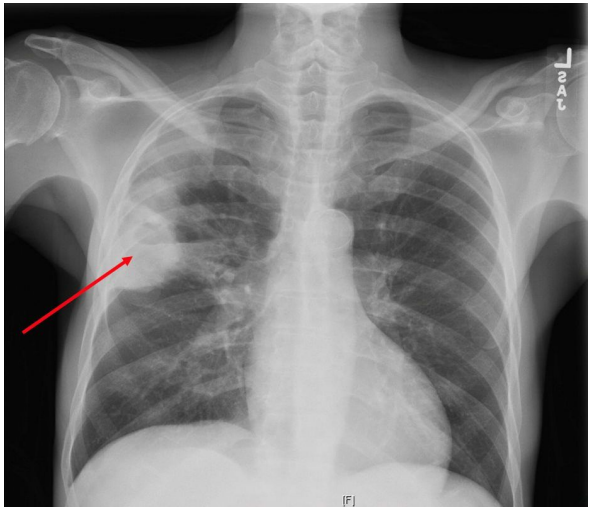


Diagram of gas exchange in an alveolus

[1] American Lung Association. (2014). *Our Impact*. Lung.org; https://www.lung.org/about-us/our-impact

Our Goal:

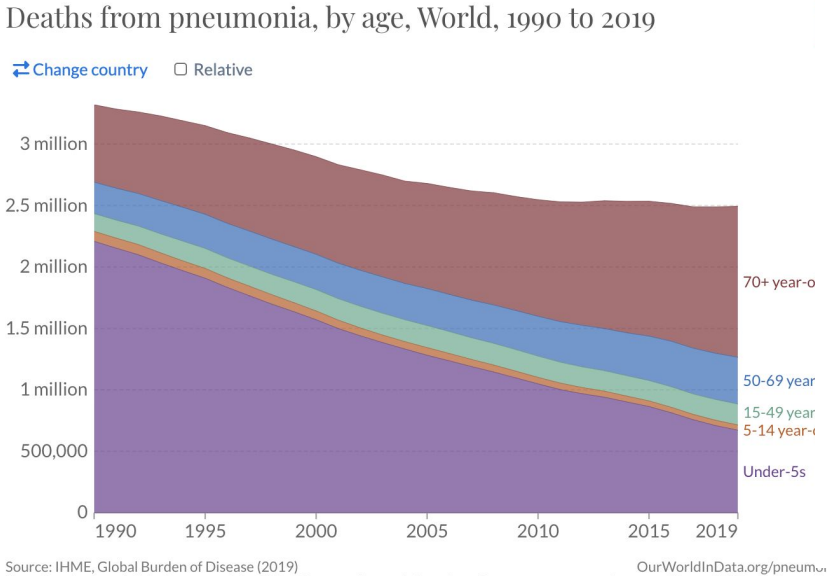
- To model gas exchange (O_2 & CO_2) in the lungs under healthy and diseased conditions with pneumonia

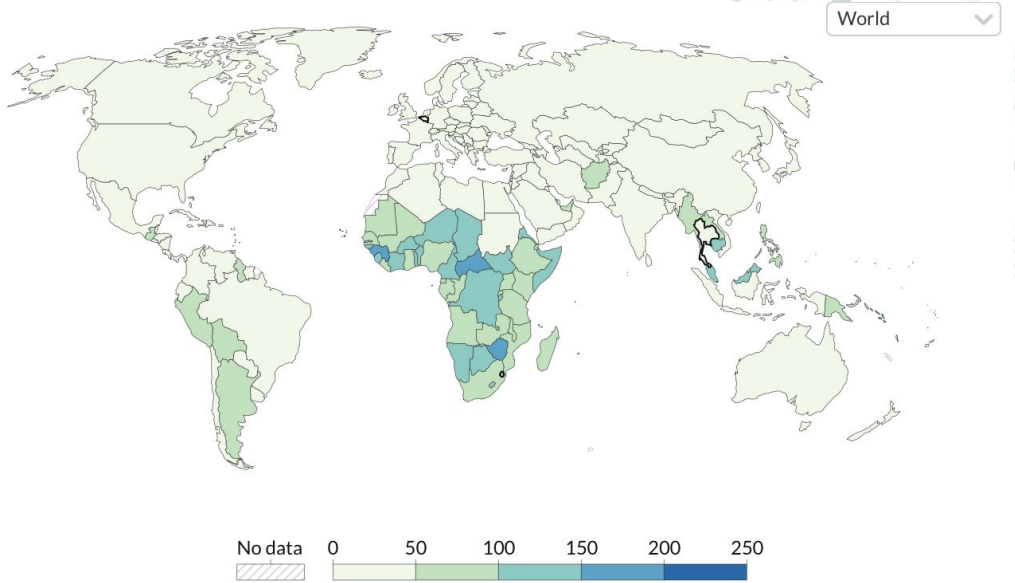


Lung with pneumonia

[3] Cannella, Joseph, and Christina LeBedis. Right Upper Lobe Pneumonia. *The Common Vein*, https://lungs.thecommonvein.net/pneumonia/

Motivation: Why Pneumonia?

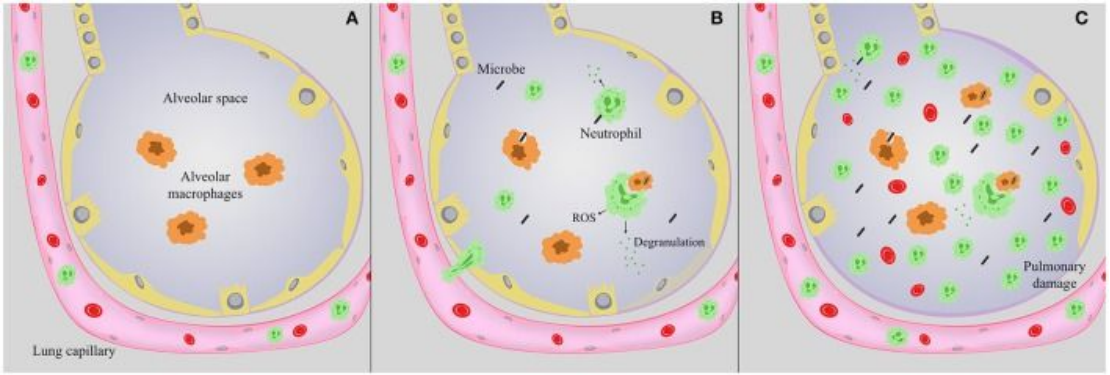




[4] Dadonaite, Bernadeta, and Max Roser. (2018). Pneumonia. *Our World in Data*, https://ourworldindata.org/pneumonia.

Physiology of Pneumonia

1. Alveolar macrophages recruit neutrophils from pulmonary vasculature
2. Neutrophils release degrading toxins
3. Toxins harm alveolar tissue
4. Fluid leaks into alveolar space[2]



[7] Pechous RD (2017) With Friends Like These: The Complex Role of Neutrophils in the Progression of Severe Pneumonia. *Front. Cell. Infect. Microbiol*. 7:160. doi: 10.3389/fcimb.2017.00160

Overview of Key Results

- 40% increase in the distance of diffusion due to purulent fluid buildup
- Oxygen diffusion from alveoli to blood and carbon dioxide diffusion from blood to alveoli both decrease 40% with pneumonia
- 8.65% increase in O_2 in expired air
- 37.7% decrease in CO_2 in expired air

Chemical Constituents We Tracked:

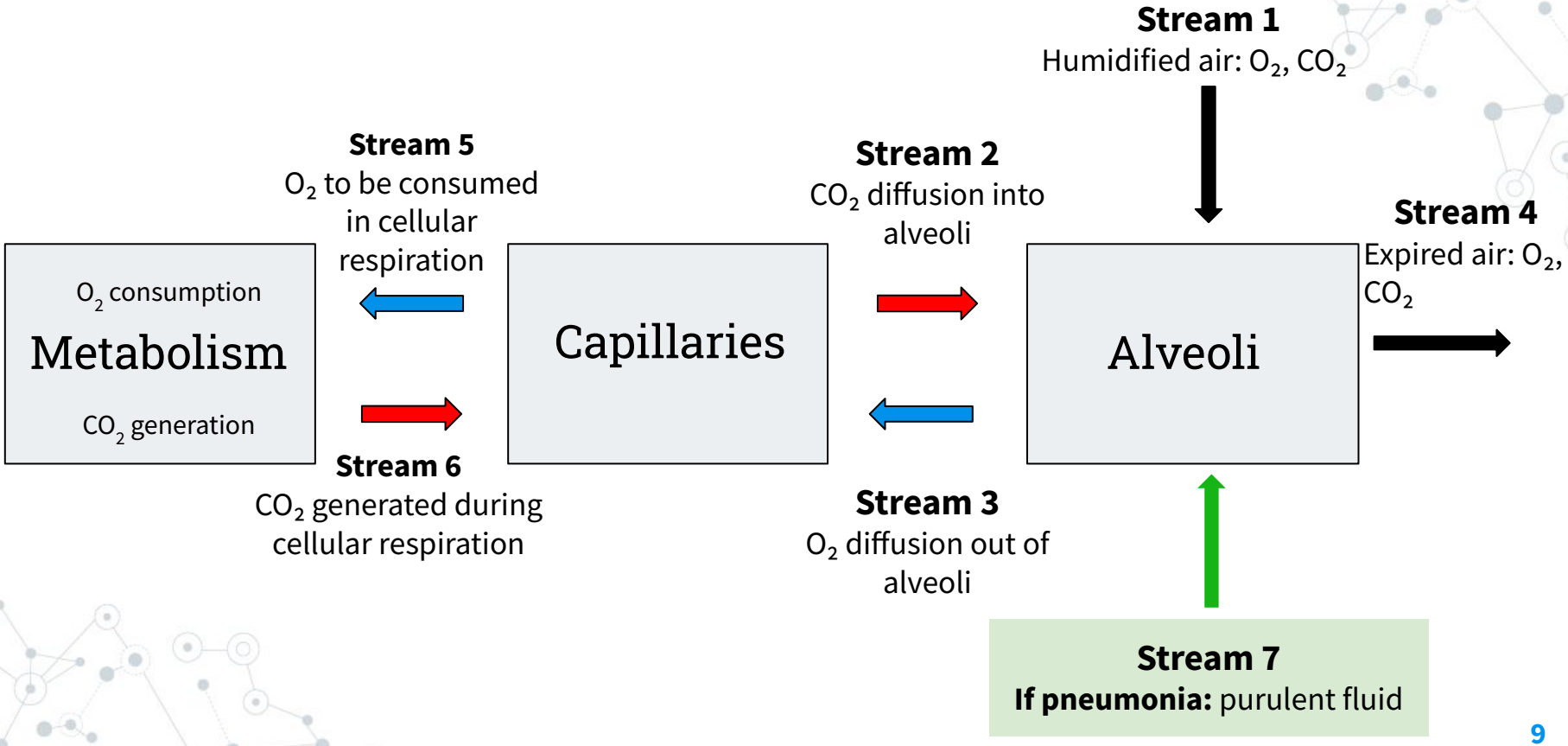
1) O_2
2) CO_2
3) Purulent Fluid: H_2O (95%) + Neutrophils

Key Assumptions

1. Healthy = steady-state ($ACC_{SYS} = 0$)
2. Pneumonia = dynamic ($ACC_{SYS} \neq 0$)
   a. ACC is only due to purulent fluid
3. Streams 2 and 3 are unidirectional: O_2 out and CO_2 in
4. Distance is the only factor changing the rates of diffusion during pneumonia
5. Metabolism does not change with pneumonia

Overall Model

Blue = O_2
Red = CO_2



Key Mathematical Equations

Diffusion:

$$D \propto \frac{\Delta P \times A \times S}{d \times \sqrt{MW}} \qquad \dot{n} \propto \frac{k}{d}$$

Overall System:

Healthy

$$\dot{n}_1 = \dot{n}_4$$

Diseased

$$\dot{n}_1 + \dot{n}_7 = \dot{n}_4$$

$\dot{n}_s$ = molar flow rate, stream *s*
D = diffusion rate
d = diffusion distance
ΔP = partial pressure difference
A = total surface area of diffusing surface
S = solubility
MW = molecular weight

Ideal Gas Law

$$PV = nRT$$

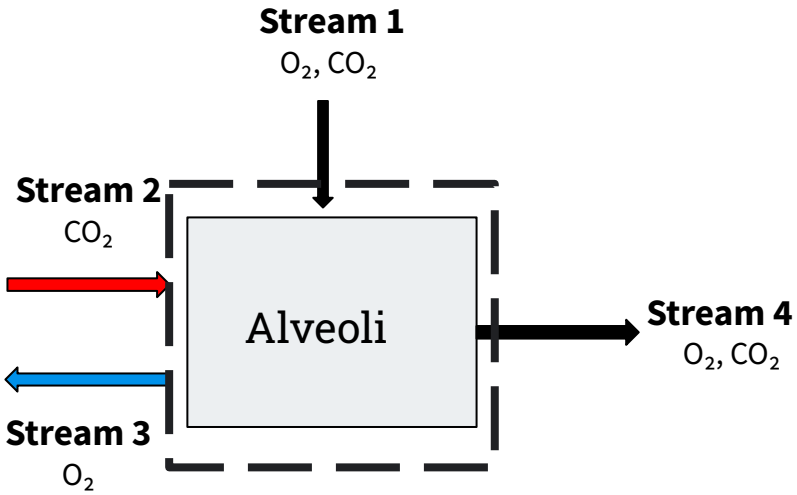
- Used to calculate molar flow rates of inlet (humidified) and outlet (expired) streams

P = partial pressure difference of gas between alveoli and pulmonary capillaries
V = tidal volume, volume of air expired in each breath
n = moles of air in each breath
R = gas constant
T = body temperature

$$n = \frac{(1\ \text{atm})(0.5\ \text{L/breath})}{\left(0.08206\ \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}}\right)(310\ \text{K})}$$

$$n = 0.01966\ \text{mol} \,/\, 1\ \text{Breath}$$

Alveoli Unit in a Healthy State



Blue = O_2
Red = CO_2

Total Mole Balance:

$$\dot{n}_1 + \dot{n}_2 = \dot{n}_3 + \dot{n}_4$$

O_2: $\dot{n}_{1,O2} = \dot{n}_3 + \dot{n}_{4,O2}$

CO_2: $\dot{n}_{1,CO2} + \dot{n}_2 = \dot{n}_{4,CO2}$

Mole Balances on Alveoli System:

$$\dot{n}_{1,O_2} = \left(0.01966 \frac{\text{mol}}{\text{breath}}\right)(0.1967)\left(12 \frac{\text{breath}}{\text{min}}\right)$$

$$= 4.64 \times 10^{-2} \frac{\text{mol}}{\text{min}}$$

$$\dot{n}_{4,O_2} = \left(0.01966 \frac{\text{mol}}{\text{breath}}\right)(.157)\left(12 \frac{\text{breath}}{\text{min}}\right)$$

$$= 3.70 \times 10^{-2} \frac{\text{mol}}{\text{min}}$$

$$\dot{n}_{1,CO_2} = \left(0.01966 \frac{\text{mol}}{\text{breath}}\right)(0.0004)\left(12 \frac{\text{breath}}{\text{min}}\right)$$

$$= 9.44 \times 10^{-5} \frac{\text{mol}}{\text{min}}$$

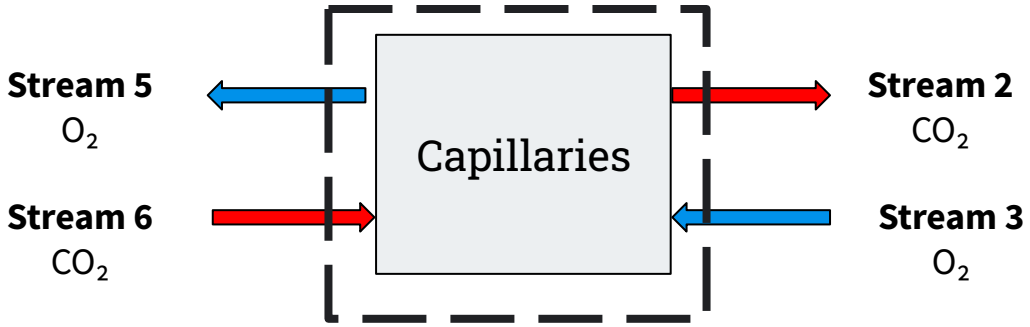
$$\dot{n}_{4,CO_2} = \left(0.01966 \frac{\text{mol}}{\text{breath}}\right)(0.036)\left(12 \frac{\text{breath}}{\text{min}}\right)$$

$$= 8.49 \times 10^{-3} \frac{\text{mol}}{\text{min}}$$

$$\dot{n}_{3,O_2} = \dot{n}_{1,O_2} - \dot{n}_{4,O_2} = 9.40 \times 10^{-3} \frac{\text{mol}}{\text{min}}$$

$$\dot{n}_{2,CO_2} = \dot{n}_{4,CO_2} - \dot{n}_{1,CO_2} = 8.40 \times 10^{-3} \frac{\text{mol}}{\text{min}}$$

Capillaries Unit in a Healthy State



***ASSUME:** Steady State (ACC = 0)

Blue = O_2
Red = CO_2

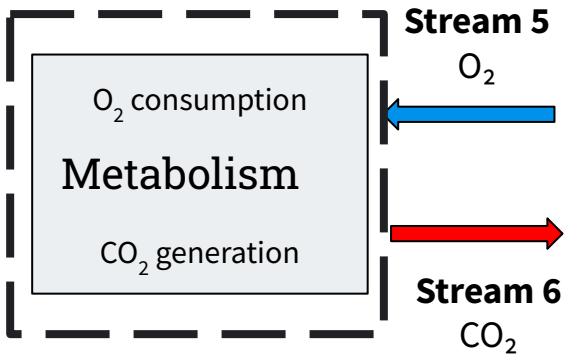
Total Mole Balance:

$$\dot{n}_3 + \dot{n}_6 = \dot{n}_2 + \dot{n}_5$$

O_2: $\dot{n}_3 = \dot{n}_5 = 9.4\times10^{-3}$ mol/min

CO_2: $\dot{n}_6 = \dot{n}_2 = 8.4\times10^{-3}$ mol/min

Metabolism Unit in a Healthy State



Blue = O_2
Red = CO_2

Total Mole Balance:

$\dot{n}_5 = \dot{n}_6$

9.4×10^{-3} mol/min ≃ 8.4×10^{-3} mol/min

***ASSUME:** 100% consumption of Oxygen and conversion to Carbon Dioxide in Metabolism

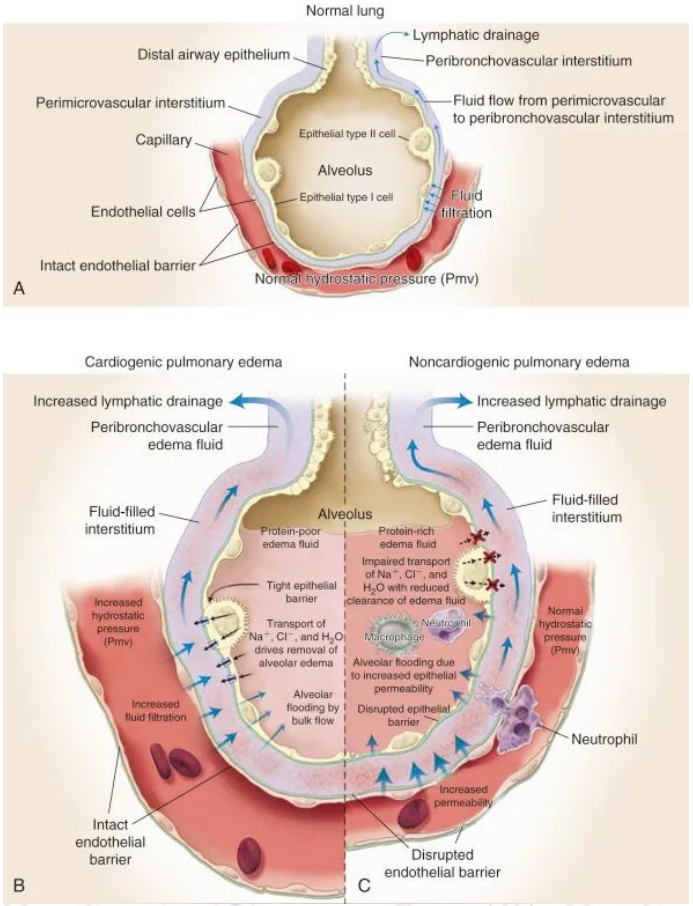
Modeling Pneumonia

Accumulation of fluid causes:

- Increased diffusion distance → decreased O_2 and CO_2 exchange w/ capillaries
- Increased Exhalation of O_2 and CO_2
- Decreased O_2 in blood

ASSUME:

- “Severe” condition takes 8 days to progress
- 60% blood oxygen saturation at “severe”



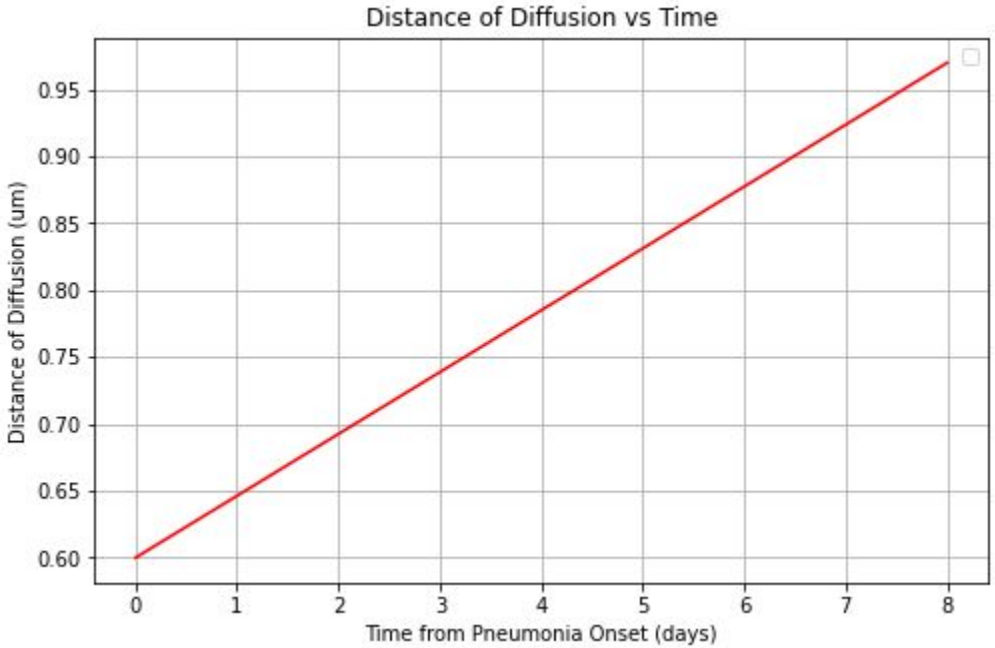
Pulmonary Edema. *Thoracic Key*, 3 July 2019, https://thoracickey.com/pulmonary-edema-4/

Healthy vs. Extreme Pneumonia

$$w_{O2,\,diseased} = \frac{0.6}{0.97} w_{O2,\,healthy} \qquad d_{diseased} = \frac{0.97}{0.6} d_{healthy}$$

	Healthy Case	Extreme Pneumonic Case
Oxygen Blood Saturation	97%	60%
Distance of Diffusion	0.6 μm *thickness of pulmonary membrane	0.97 μm *thickness + purulent fluid

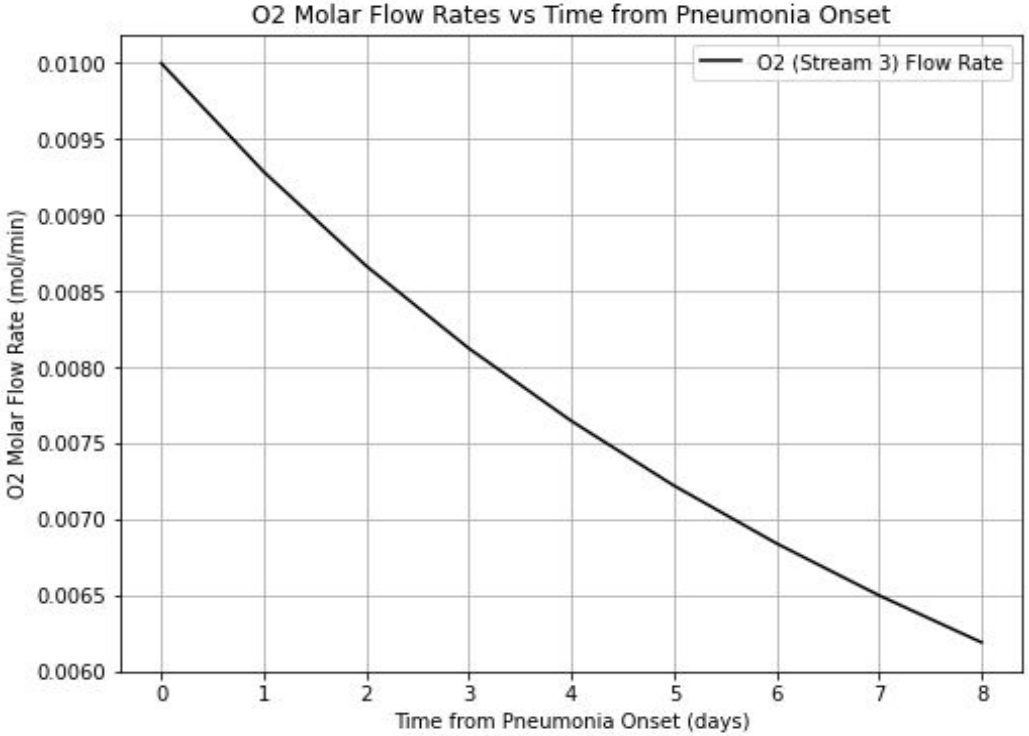
Results: Change in Diffusion Distance as a Function of Pneumonia Onset

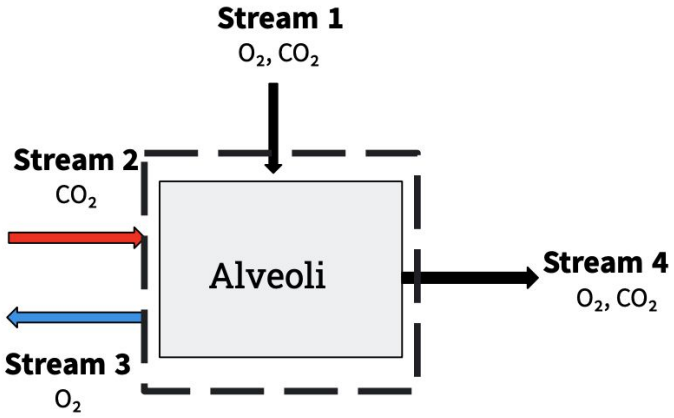


$$d(t) = 0.6\ \mu m + \left(\frac{(0.97 - 0.6)\ \mu m}{(8 - 0)\ days}\right)t$$

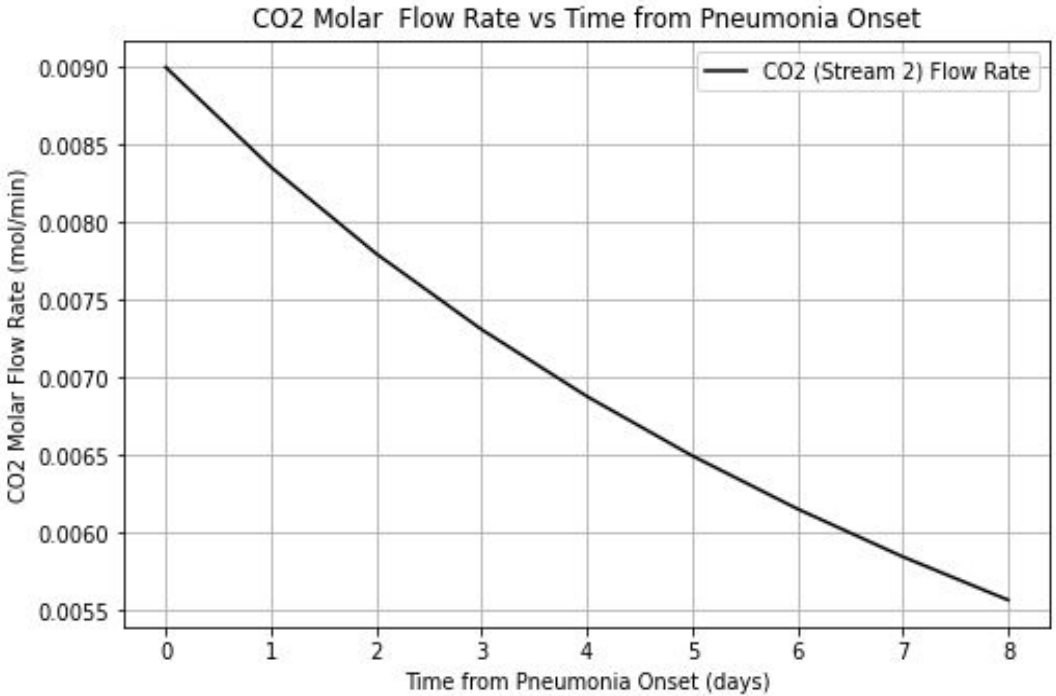
- 8 days for pneumonia to progress to severe
- Healthy distance ~ 0.6 µm
- Diseased distance ~ 0.97 µm

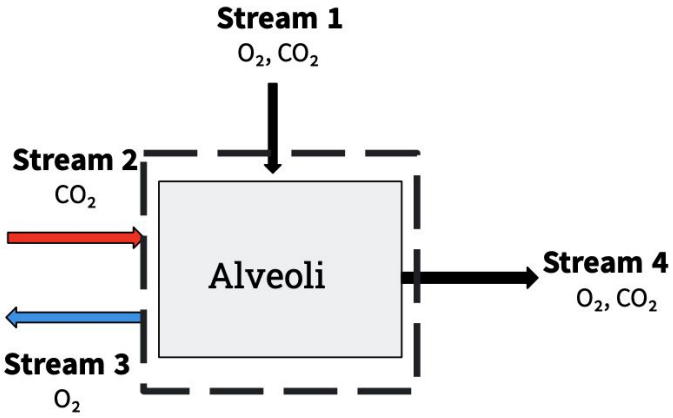
Results: Change in O_2 Diffusion as a Function of Pneumonia Onset



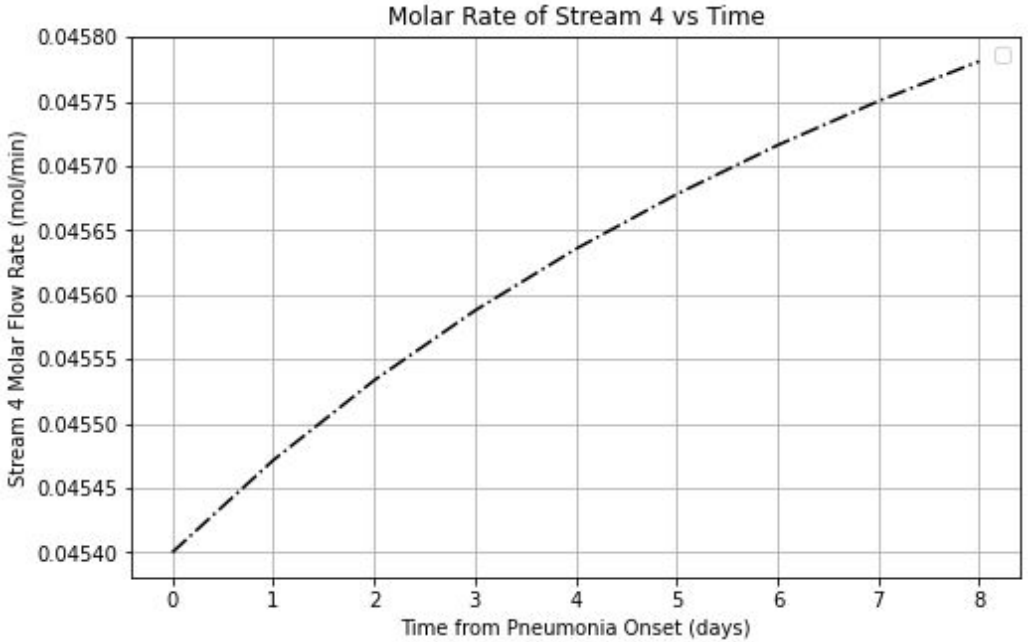


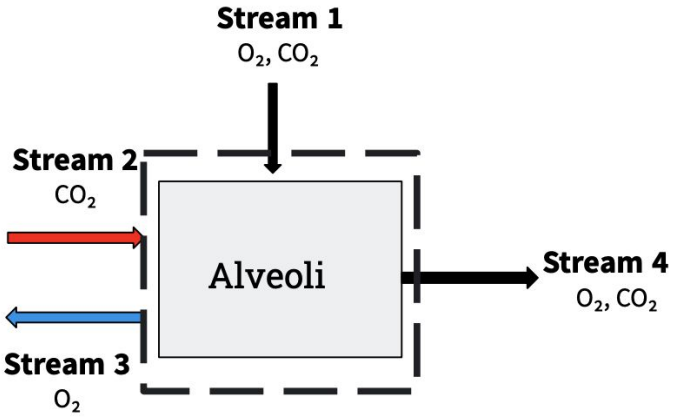
Results: Change in CO_2 Diffusion as a Function of Pneumonia Onset



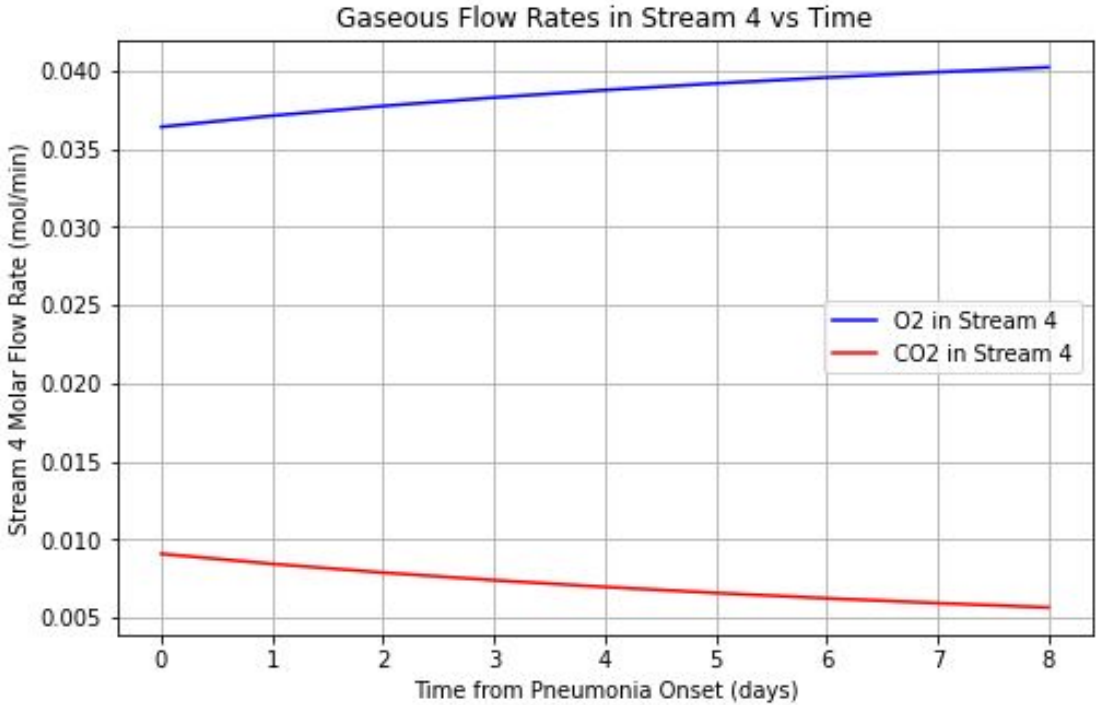


Results: Change in Molar Flow Rate of Expired Air as a Function of Pneumonia Onset





Results: Change in Molar Flow Rate of Expired O_2 and CO_2 as a Function of Pneumonia Onset



Calculation of Expired O_2 and CO_2 Molar Flow Rates at Severe Pneumonic Case

Severe pneumonia (8 days):

O_2

$$\dot{n}_{3,f} = \left(\frac{0.6}{0.97}\right)\dot{n}_{3,i} = \left(\frac{0.6}{0.97}\right)(9.4\cdot10^{-3}) = 5.81\cdot10^{-3}\ \text{mol/min}$$

$$\dot{n}_{1,O_2 f} = \dot{n}_{1,O_2 i} = 4.6\cdot10^{-2}\ \text{mol/min}$$

$$\dot{n}_{4,O_2 f} = \dot{n}_{1,O_2 f} - \dot{n}_{3,f} = 4.6\cdot10^{-2} - 5.81\cdot10^{-3} = 4.02\cdot10^{-2}\ \text{mol/min}$$

$\dot{n}_{4,O_2,i} = 3.7\cdot10^{-2}$ mol/min $\Rightarrow \dot{n}_{4,O_2 f} > \dot{n}_{4,O_2 i} \Rightarrow$ amount of O_2 in expired air ↑

CO_2

$$\dot{n}_{2f} = \left(\frac{0.6}{0.97}\right)\dot{n}_{2,i} = \left(\frac{0.6}{0.97}\right)(8.4\cdot10^{-3}) = 5.20\cdot10^{-3}\ \text{mol/min}$$

$$\dot{n}_{1,CO_2 f} = \dot{n}_{1,CO_2 i} = 9.44\cdot10^{-5}\ \text{mol/min}$$

$$\dot{n}_{4,CO_2 f} = \dot{n}_{1,O_2 f} + \dot{n}_{2,f} = 9.44\cdot10^{-5} + 5.20\cdot10^{-3} = 5.29\cdot10^{-3}\ \text{mol/min}$$

$\dot{n}_{4,CO_2 i} = 8.49\cdot10^{-3}$ mol/min $\Rightarrow \dot{n}_{4,CO_2 f} > \dot{n}_{4,CO_2 i} \Rightarrow$ amount of CO_2 in expired air ↓

Results: Change in O_2 and CO_2 Molar Flow Rates Expired at Severe Pneumonic Case

	Molar Flow Rate in Expired Air @ t = 0 days (mol/min)	Molar Flow Rate in Expired Air @ t = 8 days	% Difference
O_2	3.70×10^{-2}	4.02×10^{-2}	8.65% increase
CO_2	8.49×10^{-3}	5.29×10^{-3}	37.7% decrease

Weaknesses

- Assumes partial pressure difference remains constant, when it realistically varies
- Linear interpolation of disease progression
- Ignored dead space air in expired air stream
- Capillaries and metabolism units are usually dynamic

Strengths

- Our numerical results closely resemble literature values

	Literature Value	**Calculated Value**
O_2 Diffusion Rate	1.1×10^{-2} mol/min	1.0×10^{-2} mol/min
CO_2 Diffusion Rate	9.0×10^{-2} mol/min	8.0×10^{-2} mol/min

Strengths (continued)

Consumption rate of O_2 = 250 ml/min

$$250 \frac{ml}{min}\left(\frac{1\,mol}{32\,g\,O_2}\right)\left(\frac{1.43\,g}{1\,L}\right)\left(\frac{1\,L}{1000\,ml}\right) = 0.0112 \frac{mol}{min}$$

CHECK: $\dot{n}_{3,O_2} = 0.0094 \frac{mol}{min}$

Molar rate of O_2 diffusion $\approx 0.01 \frac{mol}{min}$

Production rate of CO_2 = 200 ml/min

$$200 \frac{ml}{min}\left(\frac{1\,mol}{44\,g\,(CO_2)}\right)\left(\frac{0.00197\,g}{1\,ml}\right) = 0.00895 \frac{mol}{min}$$

CHECK: $\dot{n}_{2,CO_2} = 0.0084 \frac{mol}{min}$

Molar rate of CO_2 diffusion $\approx 0.009 \frac{mol}{min}$

Strengths (continued)

- Models the expected impairment of gas exchange under pneumonic conditions
- Simple model to understand lung functionality in healthy vs diseased states
  - Ability to add complexity for better accuracy
  - Can apply to other respiratory illnesses (COVID-19)

Implications of Our Model

Decreased oxygen and increased carbon dioxide levels in the bloodstream ⇒ hypoxemia and hypercapnia

- Risk of organ failure, trouble breathing, heart and brain malfunction

Model could be improved by considering Metabolism:

- Oxygen deficient tissues have slower metabolic processes and are at risk for hypoxia

Acknowledgements

Thank you to Dr. Saterbak, who mentored us greatly throughout these 6 weeks, as well as the BME 260 Teaching Assistants.

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